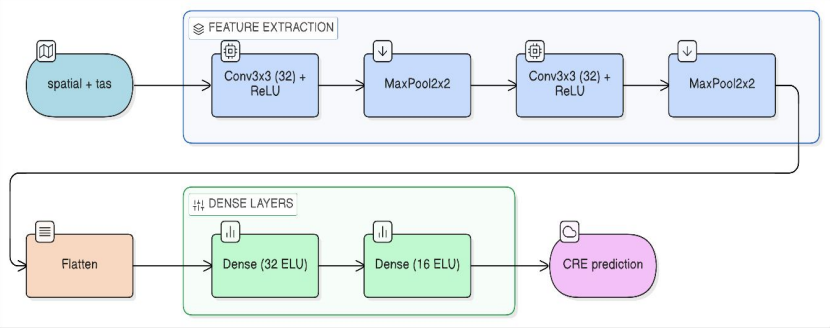
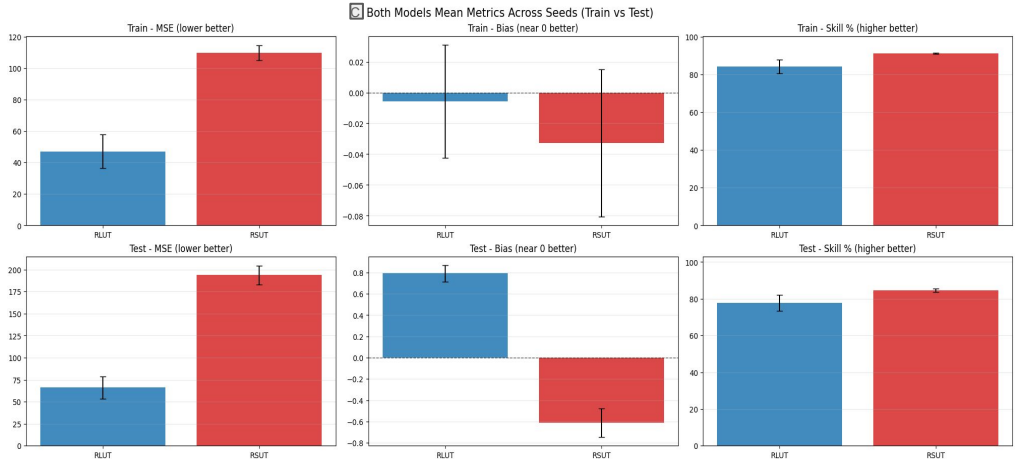
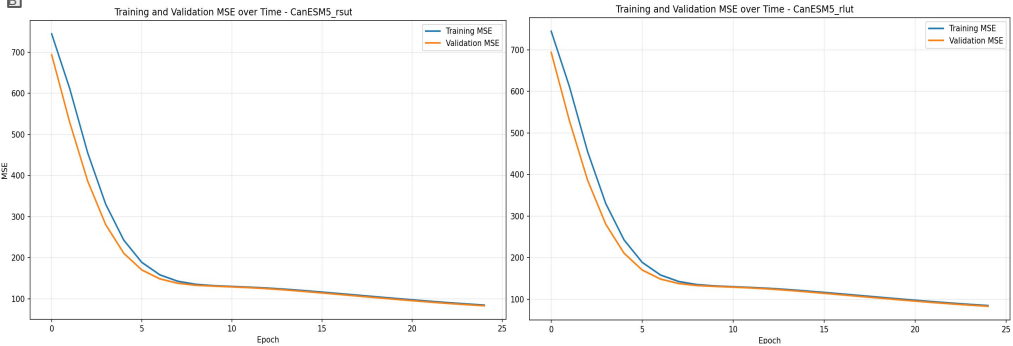


A CNN Architecture



B



D RNG Seed Variability: MSE, Bias, and Skill Score

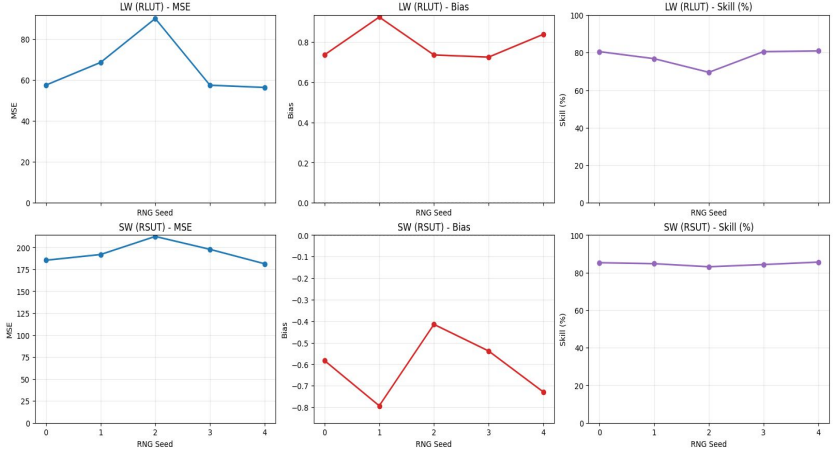


Figure 1: Model Architecture and Evaluation

(A) CNN Architecture: Shows the structure and process of the Convolutional Neural Network. The model takes near surface temperature (tas) as input, these pass through two 3x3 Convolutional layers using ReLU activation creating 32 feature maps each, then a 2x2 Max Pooling layers reduces dimensionality. The final feature maps are then flattened and passed through the fully connected layer with 32-neuron and 16-neuron using ELU activation these then output the Cloud Radiative Effect (CRE) prediction. While the ELU activation is used to improve convergence and prevent dead neurons, it may be one culprit of the overfitting we see later.

(B) Training and Validation Loss over Time: Mean Squared Error curves for both the shortwave (rsut) and longwave (rlut) models for the first random seed over time. The graphs show both the training and validation MSE across the first 25 epochs. Both curves initially show a fast decline and then a slow convergence, ensuring that the model will be able learn the connection between temperature and CRE without already overfitting in the beginning steps.

(C) Both Models Mean Metrics (Train vs Test): 3 bar charts comparing the training and testing performance as well as between longwave and shortwave models. Unfortunately, we do find overfitting as testing MSE is over 50% higher than training MSE and bias is around 0 for training vs 0.8 and -0.6 for longwave and shortwave respectively. Looking at skill score, both models still to much better than baseline and do show the ability to learn the relationship. Finally, we can see a clear comparison as longwave predictions have much lower MSE compared to shortwave, which will be explored further,

(D) RNG Seed Variability: 3 line graphs showing the variability of the model's performance across 5 different random seeds. By plotting the differences in MSE, Bias, and Skill Score for both longwave and shortwave models across different initial weights, it shows the stability of the CNN as this randomness does not dramatically change the resulting model.